

Project Proposal

Materna AI

Presented by

Joya Rani Saha, Takia Yasmin

Department of Internet of Things and Robotics
Engineering (IRE)
University of Frontier Technology, Bangladesh (UFTB)

Supervised by

Mostafiz Ahammed

Lecturer

Department of IRE,

Farzana Akter

Assistant Professor,

Department of IRE

University of Frontier Technology, Bangladesh

PROJECT TITLE:An Intelligent IoT and AI-Based Maternal Health Monitoring, Risk Prediction and Emergency Alert System

1. Introduction

Maternal health requires continuous monitoring because significant physiological changes can occur during pregnancy, while conventional healthcare mainly depends on periodic clinical visits. This can create challenges for pregnant women who need regular monitoring, particularly in rural and low-resource settings where access to healthcare facilities may be limited.

The development of Internet of Things (IoT), machine learning, wearable sensors, cloud computing, and computer vision provides opportunities for continuous and remote maternal healthcare monitoring. IoT-enabled sensors can collect physiological and environmental data and transmit them to remote monitoring platforms, while machine learning can analyze maternal health parameters to identify different levels of health risk.

However, maternal safety is not limited to physiological conditions. Events such as falls or sudden abnormal movements may also require immediate assistance. Existing approaches often address physiological monitoring, risk prediction, or fall detection separately. Therefore, there is a need for an integrated system that combines real-time health monitoring, AI-based risk prediction, fall detection, GPS tracking, emergency alerts, and remote healthcare access within a single platform.

To address this need, Materna AI proposes an integrated IoT and AI-based maternal healthcare system. The system uses an ESP32-based sensing unit to collect physiological, motion, environmental, and location data, while AI models are used for maternal health risk prediction and fall detection. The system also incorporates GPS tracking, emergency alerts, cloud monitoring, and a Flask-based web dashboard for remote monitoring.

2. Problem and Proposed Solution

Key Problems

- ✓ Pregnant women may require continuous health monitoring, but conventional healthcare mainly depends on periodic clinical visits.
- ✓ Continuous monitoring can be difficult for women living in rural and low-resource settings.
- ✓ Existing systems often focus separately on physiological monitoring, maternal risk prediction, or fall detection.

- ✓ Separate systems may not provide a unified solution for health monitoring, emergency detection, GPS tracking, and remote healthcare support.
- ✓ Falls or sudden abnormal movements may require immediate emergency attention.

Proposed Solution

To address these challenges, Materna AI integrates the following functions into a single IoT and AI-based platform:

- ✓ Real-time maternal health monitoring using physiological and environmental sensors.
- ✓ AI-based maternal health risk prediction using a Random Forest model.
- ✓ Fall detection using the YOLOv8n computer vision model.
- ✓ GPS-based location tracking during emergency situations.
- ✓ Emergency alerts for high-risk conditions, abnormal ECG, or detected falls.
- ✓ MongoDB database for storing patient information, sensor data, and prediction results.
- ✓ ThingSpeak cloud monitoring for real-time sensor-data visualization.
- ✓ Flask-based web dashboard for remote monitoring and healthcare management.

3. Literature Review

Study	Model / Method	IoT / Data	Main Contribution	Limitation / Research Gap
Tappari et al. (2026) [1]	Random Forest, IoT	Wearable sensors, ESP32, ThingSpeak	Maternal and child monitoring	No fall detection, GPS, or doctor interaction
Karventhan et al. (2026) [2]	IoT wearable	Wearable belt, cloud	Maternal and fetal monitoring	No ML risk prediction or fall detection
Ahmed and Kashem (2020) [9]	ML, IoT	Maternal health data	Maternal risk prediction	No fall detection, GPS, or doctor interaction
Priyanka et al. (2021) [17]	IoT monitoring	BP, HR, temperature	Pregnancy vital monitoring	No ML prediction or fall detection
Assaduzzaman et al. (2023) [33]	RF, DT, XGBoost, GB	UCI maternal risk dataset	Maternal risk prediction	Static dataset; no IoT or fall detection

Proposed Materna AI (2026)	Random Forest, YOLOv8n	ESP32, ECG, MAX30105, BMP280, MPU6050, GPS, Bluetooth, Wi-Fi, ThingSpeak	Monitoring, risk prediction, fall detection, GPS, alerts, doctor interaction	Needs clinical testing, calibration, and real-world validation
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3. Objectives

The main objectives of the proposed Materna AI system are:

- To develop an IoT-based system for continuous maternal health monitoring.
- To collect important physiological and environmental parameters using ESP32 and connected sensors.
- To classify maternal health conditions into Low Risk, Mid Risk, and High Risk using machine learning.
- To develop a YOLO-based fall detection mechanism for identifying emergency events.
- To provide GPS-based location tracking during critical situations.
- To generate emergency alerts when high-risk conditions, abnormal ECG, or falls are detected.
- To store patient information, sensor records, and prediction results in a MongoDB database.
- To provide cloud-based monitoring through ThingSpeak.
- To develop a Flask-based web dashboard for remote monitoring and healthcare management.
- To integrate health monitoring, risk prediction, fall detection, location tracking, and emergency response into a single platform.

4. Proposed System Methodology

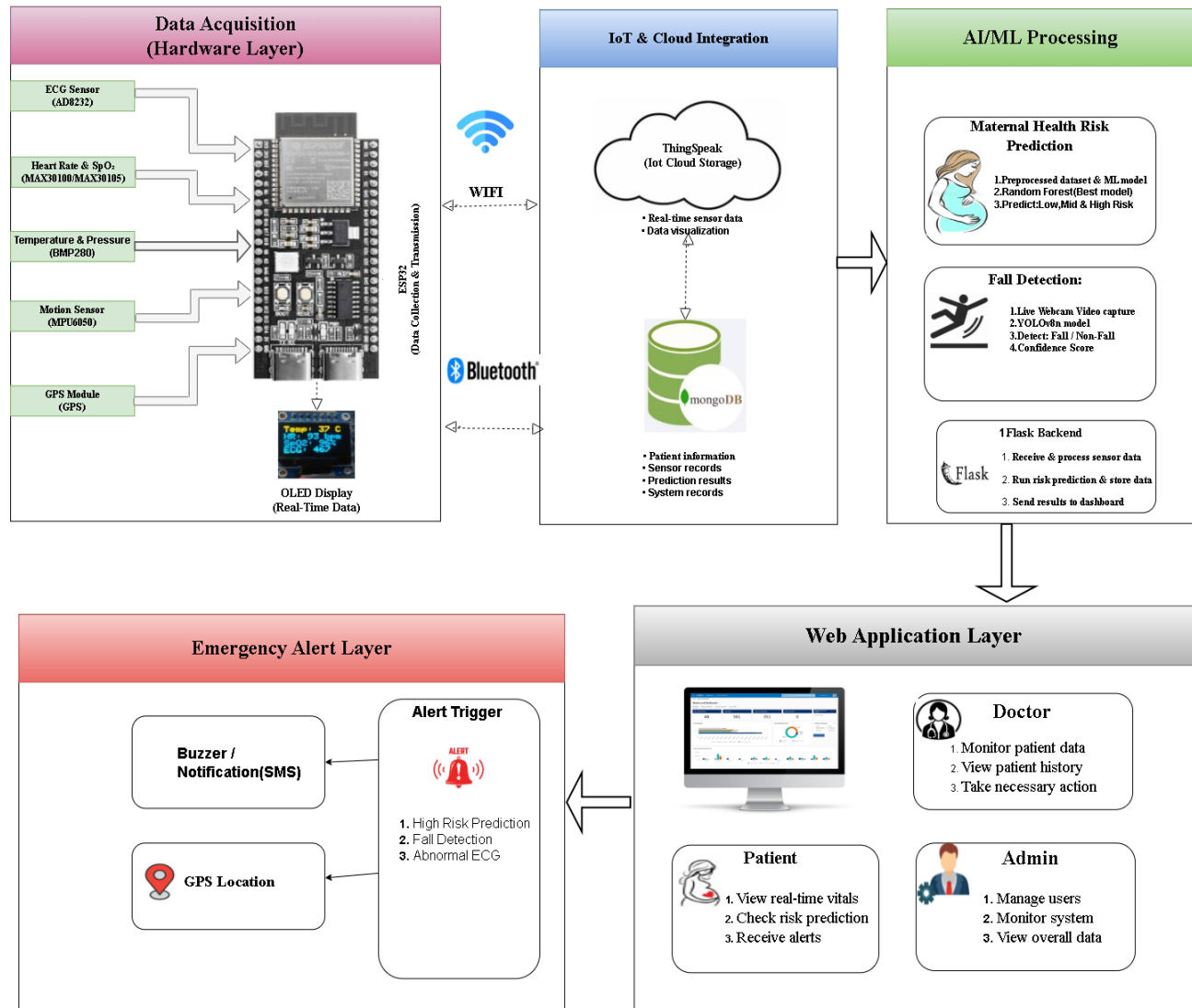


Figure 1: Overall Methodology and System Architecture of the Proposed Materna AI System

The proposed Materna AI system integrates IoT-based sensing, cloud communication, database management, AI/ML processing, web-based monitoring, and emergency alert mechanisms into a unified platform. The overall methodology is illustrated in Figure 1.

5. Innovation and Novelty

Aspect	Observation	Our Novelty (Materna)
Data	Most systems use limited vital signs and static data.	Multi-sensor data (ECG, heart rate, SpO ₂ , temperature, motion, GPS) for comprehensive maternal monitoring.
Integration	IoT and ML often implemented separately.	Real-time integration of IoT sensors with AI/ML for early risk prediction.
Monitoring	Focus mainly on individual health parameters.	Continuous remote monitoring with web dashboard, mobile access and emergency alerts.
Support	Limited doctor-patient interaction.	Built-in consultation, appointment management and location tracking (GPS).
Impact	Often focus on detection only.	Helps reduce maternal risks with early detection, timely intervention and better healthcare accessibility.

The proposed Materna AI system integrates multi-sensor maternal health monitoring, IoT and AI/ML processing, continuous remote monitoring, emergency support, and GPS-based location tracking within a unified platform. This integrated approach is designed to support early risk identification, timely intervention, and improved accessibility to maternal healthcare.

7. Expected Impact

The proposed **Materna AI** system is expected to provide the following impacts:

- **Early Risk Identification:** Supports early identification of maternal health risks through AI-based prediction.
- **Continuous Monitoring:** Enables continuous monitoring of important maternal health parameters using IoT sensors.
- **Fall and Emergency Detection:** Helps identify falls and other critical conditions that may require immediate attention.
- **Rapid Emergency Response:** Provides emergency alerts along with GPS location information during critical situations.
- **Remote Healthcare Support:** Enables patients and healthcare providers to access relevant health information through a web-based platform.
- **Improved Healthcare Accessibility:** Can support maternal monitoring in rural and low-resource settings where continuous healthcare access may be limited.
- **Integrated Healthcare Platform:** Combines monitoring, risk prediction, fall detection, GPS tracking, and emergency support within a single system.

8. Hardware and Software

Hardware Components

Component	Function
ESP32	Main microcontroller for sensor data acquisition and wireless communication
ECG Sensor (AD8232)	Measures the electrical activity of the heart
MAX30105	Measures heart rate and SpO ₂
MPU6050	Detects motion and acceleration for movement monitoring
BMP280	Measures temperature and pressure
GPS Module	Provides the patient's geographical location
SSD1306 OLED	Displays heart rate, SpO ₂ , ECG, temperature, pressure, and system status
Buzzer	Provides an audible alert during emergency conditions
Bluetooth Module	Transmits ECG data and GPS location

Software / Technologies

Software / Technology	Purpose
Python	Data processing and machine learning model development
Flask	Web-based monitoring dashboard development
MongoDB	Stores patient information, sensor data, prediction results, and related system records
Ultralytics YOLO	YOLO model training and fall detection
OpenCV	Live webcam capture and video frame processing
ThingSpeak	Cloud-based sensor data monitoring
Arduino IDE	ESP32 programming and sensor integration
VS Code	Software development and project implementation

9. Limitations

1. Sensor Accuracy and Calibration

Sensor readings may be affected by noise, movement, and improper sensor placement. Regular calibration is required to ensure reliable measurements.

2. Limited Prototype Testing

The system has been tested mainly in a controlled environment. More extensive real-world testing with pregnant women is required to evaluate its practical performance.

3. Internet Dependency

Remote monitoring and cloud-based data transmission require a stable internet connection. Network interruptions may affect real-time data transmission and monitoring.

4. Data Security and Privacy

Patient health information requires strong security and privacy protection to prevent unauthorized access and ensure safe handling of sensitive data.

10. Future Works

1. Advanced AI-Based Risk Prediction

Improve the machine learning model using larger clinical datasets to achieve more accurate maternal health risk prediction.

2. Improved Sensor Performance

Apply advanced filtering and calibration techniques to improve the accuracy and reliability of sensor readings.

3. Enhanced Data Security

Implement advanced encryption, secure authentication, and access control to protect sensitive maternal health information.

4. Clinical Validation and Field Testing

Conduct extensive real-world testing with healthcare professionals and pregnant women to evaluate system accuracy, reliability, and practical usability.

11. Conclusion

The proposed Materna AI system presents an integrated approach to maternal healthcare by combining IoT-based health monitoring, AI-driven risk prediction, fall detection, GPS tracking, emergency alerts, cloud monitoring, and web-based healthcare support within a single platform.

The system uses an ESP32-based sensing unit to collect maternal health and movement data, while Random Forest is used for maternal health risk prediction and YOLOv8n is used for fall

detection. The integration of MongoDB, ThingSpeak, Flask, and GPS further supports data storage, remote monitoring, and emergency response.

Overall, Materna AI demonstrates the potential of combining IoT and Artificial Intelligence to support continuous maternal health monitoring and early risk identification. With further sensor calibration, larger clinical datasets, enhanced security, and real-world clinical validation, the system can be further developed toward a more reliable and practical maternal healthcare solution.